U.S. Provisional Patent Application — Public Filing Notice

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Application Details

Title of Invention: SYSTEM AND METHOD FOR HUMAN-APPROVED DETERMINISTIC FINALITY OF PROBABILISTIC AGENT ACTIONS VIA CONSENSUS GATING, SHAPE-SUM VERIFICATION, AND POST-QUANTUM COMMITMENT
Application Number: 64/174,202 (U.S. Provisional Application)
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Applicant: Same as inventor (inventor-owned, not assigned at filing)

Abstract

This invention addresses the gap between probabilistic AI agent behavior and deterministic business requirements. It provides a governance envelope in which actions proposed by probabilistic agents attain deterministic finality only after passing: (1) consensus gating across independent verification paths; (2) shape-sum verification of the action's structural fingerprint; and (3) post-quantum cryptographic commitment using approved PQC algorithms; with a human-approval step required before finality for operations of elevated consequence. The result is a system and method by which non-deterministic intelligence can operate inside auditable, deterministic control boundaries, with a verifiable audit trail for every committed action.

Related Public Materials

• Master thesis monograph: Deterministic Containment of Probabilistic Intelligence — DOI 10.5281/zenodo.23303240
• Dual Mesh Benchmark paper — DOI 10.5281/zenodo.23303081
• Public GitHub ledger: https://github.com/LLong2026

SQUIRL OS RESEARCH DISCLOSURE — READ FIRST

This record is part of the Squirrel OS research corpus. Open records point at the public GitHub ledger for implementation material. This disclosure is released under the patent-then-publish doctrine: the corresponding U.S. provisional application was filed and the priority date secured before this public release.

This software is a prototype and is provided for educational and research purposes only. It is not intended for production use, commercial deployment, or safety-critical environments. All systems are experimental and may contain defects on them.